

Paragliding Intelligence Platform

Volume 17 - Design System & Front-End Engineering v2 Technical

SkyThermal: Windy/Ventusky-class map experience with paragliding-first decision intelligence

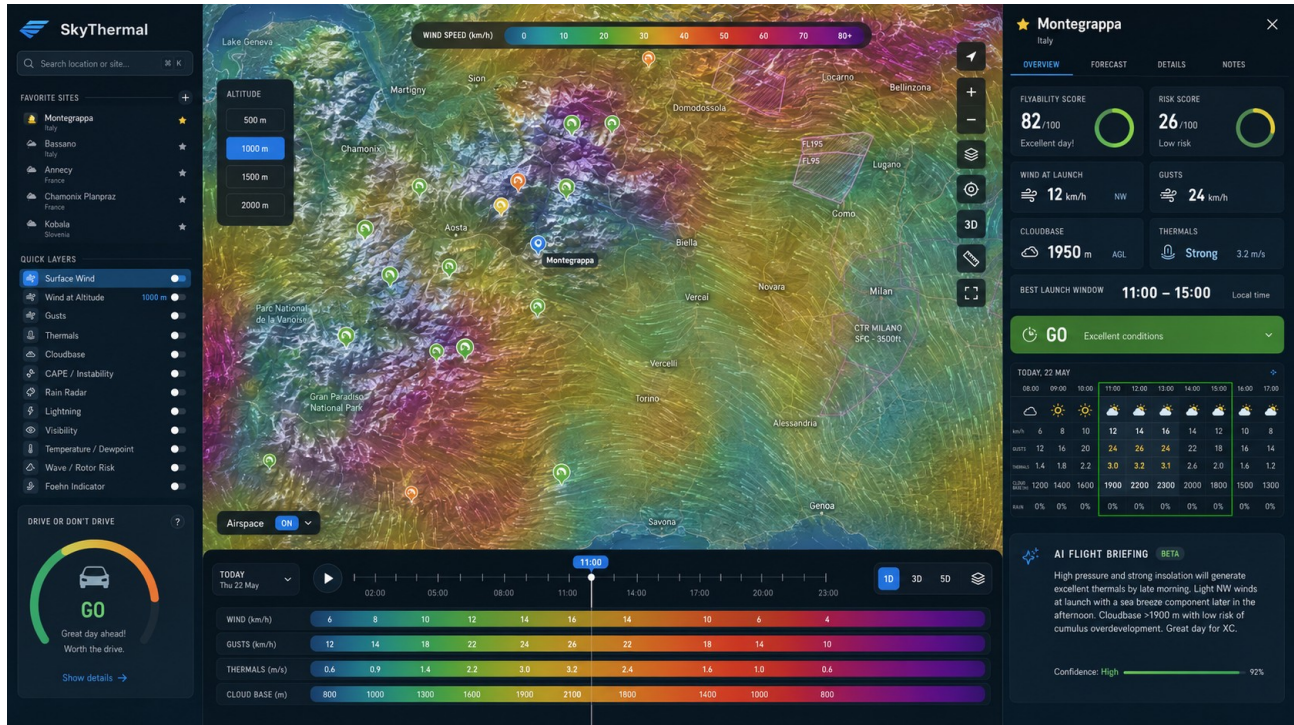


Figure 17.1 - Conceptual SkyThermal desktop map layout with weather layers, launch decision panel and timeline.

Document status: Technical design baseline. Audience: front-end engineers, mobile engineers, designers, product owners, QA and investors reviewing product readiness.

1. Purpose and Scope

This volume converts the product and architecture decisions from Volumes 01-16 into a concrete front-end engineering and design-system specification. It defines how SkyThermal should look, feel, behave, render weather data, support paid tiers, and remain usable in field conditions where paragliding pilots may be outdoors, under stress, offline, wearing gloves, or quickly checking launch safety before flying.

The goal is not to clone Windy or Ventusky. The goal is to match their level of visual confidence for animated weather maps while adding a paragliding-specific decision layer: launch compatibility, altitude wind, gust risk, thermals, cloudbase, rotor/Foehn, drive-or-do-not-drive, GO/MAYBE/NO-GO, live reality reports and AI briefings grounded in forecast data.

- Define product-wide design tokens, colors, typography, spacing and component behavior.
- Specify desktop, tablet and mobile layout behavior for map-first meteorological workflows.
- Define the map rendering architecture using MapLibre for base/vector maps and optional deck.gl/WebGL overlays for high-volume weather visualizations.
- Define layer controls for surface wind, altitude wind, gusts, thermals, cloudbase, CAPE, rain radar, lightning, visibility, temperature/dewpoint, pressure trend, wave/rotor and Foehn.
- Define UI components, props, states, accessibility rules, responsive breakpoints and implementation backlog.

2. Front-End Technical Baseline

Area	Recommended baseline	Reason
Web app shell	Next.js or Vite React with TypeScript	Fast iteration, strong ecosystem, easy integration with map rendering, API clients and design-system components.
Mobile app	Flutter for primary cross-platform implementation, or React Native if the team standardizes around React	Paragliding pilots need iOS/Android parity, offline maps, location permissions and field-grade UI consistency.
Map engine	MapLibre GL JS / MapLibre Native	Open-source WebGL vector map renderer, style-spec compatible and suitable for custom aviation/weather styling.
Advanced overlays	deck.gl or custom WebGL layer	Needed for particles, gridded weather rasters, point clouds, thermal fields and high-volume station/track visualization.
Design language	SkyThermal Design System derived from Material 3 principles but optimized for aviation/weather dashboards	Material 3 provides a proven component vocabulary, while SkyThermal requires darker map-first panels, dense data cards and safety states.
State management	URL state + server cache + local UI store	Weather maps require shareable URLs, cached forecast calls and local map/layer controls.
Offline storage	IndexedDB on web, SQLite/Hive/Isar on mobile	Needed for cached sites, forecasts, offline maps, user settings and in-flight safety state.
Testing	Storybook, Playwright, unit tests, visual regression, device labs	UI regressions can create safety confusion, especially around GO/MAYBE/NO-GO states.

External implementation references used for this volume: MapLibre GL JS documentation, deck.gl MapLibre integration, Flutter adaptive/responsive design guidance, and Material Design 3 foundations/components.

3. Information Architecture

SkyThermal is a map-first application. The map is always the primary surface. Sidebars, sheets, drawers and cards are secondary controls layered around the map. The information architecture must minimize navigation depth because pilots often need an answer in seconds.

Primary area	Desktop behavior	Mobile behavior	Critical data
Weather map	Full central viewport with overlays, wind particles, terrain and launch markers.	Full screen map with bottom sheet, layer drawer and compact site card.	Layer state, viewport, selected model run, selected time, altitude.
Left navigation	Persistent sidebar: search, favorites, quick layers, drive card.	Bottom navigation + collapsible layer drawer.	Favorites, recent sites, user location, layer toggles.
Right launch panel	Persistent selected-site panel.	Bottom sheet with snap points: compact, half, full.	Flyability, risk, window, hourly forecast, AI briefing.
Timeline	Persistent bottom scrubber.	Swipeable compact timeline below map or in bottom sheet.	Time, model run, hourly variables, play/pause.
AI briefing	Right panel card or modal.	Expandable card in site bottom sheet.	Grounded summary, confidence, hard blockers.
In-flight mode	Dedicated full-screen safety layout.	Dedicated full-screen safety layout with large buttons.	GPS, altitude, glide, airspace, alerts, SOS.

Route hierarchy

```

/
  /map?site={siteId}&layer=wind_altitude&altitude=1000&time=2026-06-05T12:00Z
  /sites/{siteId}
  /briefing/{siteId}
  /track/live
  /club/{clubId}/dashboard
  /school/{schoolId}/sessions
  /event/{eventId}/weather-center
  /admin/moderation

```

4. Layout System and Breakpoints

The layout system uses a responsive shell that changes navigation density based on viewport width and input mode. Desktop prioritizes three-column productivity; tablet prioritizes map plus one panel; mobile prioritizes a map with progressive disclosure.

Breakpoint	Width	Shell mode	Panels	Target use
mobile-s	320-374 px	Map + compact bottom sheet	No permanent sidebars	Phone in field, quick checks.
mobile	375-767 px	Map + bottom sheet + layer drawer	Bottom nav, swipe sheets	Common pilot phone workflow.
tablet	768-1023 px	Map + right drawer	One persistent panel at a time	Launch planning on tablets.

desktop	1024-1439 px	Left sidebar + map + right panel	Persistent left/right panels	Planning at home, clubs/schools.
wide	1440+ px	Three-column + expanded timeline	Persistent panels, richer tables	Investor/demo/ professional dashboard.

CSS layout contract

```
.app-shell {
  display: grid;
  grid-template-columns: var(--sidebar-w) minmax(0, 1fr) var(--detail-w);
  grid-template-rows: 1fr var(--timeline-h);
}
@media (max-width: 1023px) {
  .app-shell { grid-template-columns: 1fr; grid-template-rows: 1fr; }
  .sidebar, .detail-panel { position: fixed; }
}
```

5. Design Tokens

Design tokens are product infrastructure. They prevent inconsistent weather colors, broken contrast, uncontrolled spacing and duplicate components. All web and mobile apps should use the same token names, with platform-specific exports generated by the design-system package.

Token group	Examples	Rules
Color semantic	surface, panel, panelGlass, textPrimary, textMuted, actionBlue, safetyGo, safetyMaybe, safetyNoGo, riskExtreme	Semantic colors must not be hardcoded in components.
Weather ramps	windRamp, gustRamp, thermalRamp, cloudbaseRamp, rainRamp, capeRamp, rotorRamp	Each map layer owns a scientifically meaningful color ramp plus a UI legend.
Typography	display, title, label, body, dataMono, mapLabel	Data-heavy cards use tabular numerals; map labels use high-contrast shadows.
Spacing	space-1..space-10	4 px base grid; dense mode uses 2 px exceptions for tables.
Radius	radius-sm, radius-md, radius-lg, radius-pill	Panels use radius-lg; chips use radius-pill.
Elevation	shadow-mapPanel, shadow-floating, shadow-modal	Weather panels must remain readable over bright terrain.
Motion	duration-fast, duration-map, easing-standard	Avoid flashy motion; critical alerts must animate only enough to be noticed.
Z-index	z-map, z-overlay, z-panel, z-modal, z-critical	Critical alert overlays always win over non-critical UI.

Example design-token JSON

```
{
  "color": {
    "surface": "#07111F",
    "panel": "rgba(7,18,31,0.88)",
    "actionBlue": "#2F9BFF",
    "safetyGo": "#4CAF50",
    "safetyMaybe": "#FFC107",
    "safetyNoGo": "#EF5350",
```

```

    "riskExtreme": "#9C27B0"
  },
  "weatherRamp": {
    "wind": ["#267BD4", "#1CC0B2", "#59C55A", "#D2BD32", "#F1852D", "#DD484A", "#C33190", "#732BCE"],
    "thermal": ["#1C4D9B", "#36A2EB", "#4CAF50", "#FFD54F", "#FF8A35"],
    "risk": ["#3CB371", "#FFD54F", "#FF8A35", "#EF5350", "#9C27B0"]
  }
}

```

6. Safety Color and Wording Rules

Weather apps can use decorative colors. SkyThermal cannot. Color and wording directly influence pilot decisions, so safety states must be standardized across the entire app.

State	Color	Allowed wording	Forbidden wording	Behavior
GO	Green	Good window, conditions favorable, low risk for selected profile	Guaranteed safe, safe to fly, launch now	Always include confidence and reason.
MAYBE	Yellow/amber	Marginal, verify live conditions, suitable only for specified skill profile	Probably fine, should be OK	Force expansion of risk reasons before route sharing.
NO-GO	Red	Do not fly, hard blocker detected, conditions outside limits	Bad weather only	Show blocker and alternate sites if available.
UNKNOWN	Gray/blue	Insufficient data, no recommendation	No risk detected	Used when model age, missing variables or sensor quality fail gates.
EXTREME	Purple/red	Critical hazard, land soon, emergency risk	Warning only	Highest visual priority; voice/audio in in-flight mode.

7. Component Catalog

Component	Responsibility	Core props	States
AppShell	Owns responsive layout, panel slots, map viewport and global shortcuts.	layoutMode, selectedSiteId, activeLayer, selectedTime	Desktop/tablet/mobile, loading, degraded API.
MapViewport	Renders basemap, weather layers, launch markers, airspace, tracks and hazard zones.	viewport, layers, time, altitude, modelRunId	2D/3D, low GPU, offline, no data.
LayerControlPanel	Layer toggles, altitude selector, model selector and entitlement badges.	availableLayers, enabledLayers, selectedAltitude, tier	Free locked, Pro enabled, missing data.
TimelineScrubber	Time selection, playback, model-run comparison and hourly strips.	timeRange, selectedTime, playbackRate, series	Drag, play, keyboard, disabled.
SiteDecisionPanel	Selected launch details, GO/MAYBE/NO-GO, score	site, forecast, flyability, risk, userProfile	No site, no forecast, hard blockers.

	cards and reason tree.		
ScoreGauge	Circular or bar metric for flyability/risk/confidence.	value, max, status, label	Normal, warning, error, skeleton.
WeatherMetricCard	Single metric with trend, unit and confidence.	label, value, unit, trend, confidence	Unknown, stale, sensor/live value.
AIFlightBriefingCard	Grounded summary of forecast and site playbook.	briefing, citations/provenance, confidence	Blocked, beta, requires Pro+.
DriveDecisionCard	Worth driving calculation with travel window.	origin, site, driveTime, flyWindow, score	No location, too late, maybe.
InFlightAlertBanner	Critical in-flight safety messages.	severity, message, action, expiresAt	Voice-ready, high contrast, dismiss rules.

Example TypeScript component contract

```
export type SafetyState = 'GO' | 'MAYBE' | 'NO_GO' | 'UNKNOWN' | 'EXTREME';
```

```
export interface SiteDecisionPanelProps {
  siteId: string;
  selectedTime: string;
  pilotProfileId?: string;
  forecast: SiteForecastSummary;
  flyability: FlyabilityDecision;
  entitlement: EntitlementSnapshot;
  onCreateAlert(rule: AlertRuleDraft): void;
  onStartNavigation(siteId: string): void;
}
```

8. Main Weather Map Layer Matrix

This table must exist in the PRD, Weather Engine, API Specification and Front-End Engineering documents. It is the common contract between product, backend, weather processing and UI.

Layer ID	UI name	Paragliding purpose	Rendering	Units	Tier	Front-end behavior
surface_wind	Surface wind	Launch safety	streamlines + raster ramp	km/h, direction deg	Free	Refresh with each forecast update; live station override chip.
altitude_wind	Wind at altitude: 500m, 1000m, 1500m, 2000m	Drift and XC planning	streamlines + altitude selector	km/h, direction deg, altitude m AGL/MSL	Pro	Altitude selector must be visible and URL-shareable.
gusts	Gusts	Collapse / launch danger	raster heatmap + contour	km/h	Free/Pro detail	Show gust spread and hard blocker.
wind_direction	Wind direction	Launch compatibility	arrows/barbs + launch orientation rose	degrees, cardinal	Free	Tie to launch allowed directions.
thermals	Thermals	Soaring potential	heatmap + trigger points	m/s, confidence	Pro	Trigger points unlock in Pro+.
cloudbase	Cloudbase	XC potential /	raster +	m AGL/MSL	Pro	AGL/MSL

		cloud danger	isolines			switch required.
cape_instability	CAPE / instability	Overdevelopment risk	heatmap + warning polygons	J/kg, risk class	Pro+	Must influence NO-GO when storm blockers present.
rain_radar	Rain radar	No-fly zones	animated radar raster	mm/h or dBZ	Free	Nowcast priority over model precip for current time.
lightning	Thunderstorm /lightning	Critical danger	strike symbols + storm cells	strike count, age	Free	Critical alert layer; in-flight voice warnings.
visibility	Visibility	Safe navigation	raster/ contours	km	Pro	Low visibility can hard-block beginner profiles.
temp_dewpoint	Temperature/ dewpoint	Cloud formation	dual scalar layer + meteogram	Celsius	Free	Used for cloudbase and fog risk explanation.
pressure_trend	Pressure trend	Weather change	isobars + trend arrows	hPa, hPa/3h	Pro	Good for macro weather context and Foehn analysis.
wave_rotor	Wave / rotor risk	Mountain danger	hazard polygons + ridge lee shading	risk 0-100	Pro+	Requires terrain and wind-at-altitude.
foehn	Foehn indicator	Very important in mountains	regional risk overlay	risk 0-100, pressure gradient	Pro+	Must show uncertainty and model disagreement.

9. Map Rendering Architecture

The map renderer is a multi-layer system. Base maps, terrain, labels, launch markers and airspace should be vector-tile driven. Weather overlays can be raster tiles, vector isolines, particle fields or deck.gl layers depending on the variable and zoom level.

Layer type	Technology	Data source	Caching	Notes
Basemap	MapLibre style with vector tiles	PMTiles/vector tile server	CDN + local cache	Separate aviation/weather style from generic road style.
Terrain/hillshade	MapLibre hillshade +	Copernicus/SRTM	CDN + offline packs	High relief helps

	DEM raster tiles	derived tiles		pilots understand rotor/valley risk.
Wind particles	Custom WebGL layer or raster-particle layer where supported	Forecast cube UV vectors	Tile cache by model/time/altitude	Throttle particle density by GPU capability.
Scalar weather rasters	RGBA PNG/WebP tiles or Cloud Optimized GeoTIFF preprocessing	Forecast grid	CDN and browser cache	Use consistent legends and no-data masks.
Isolines	Vector tiles / GeoJSON at lower zoom	Contour generator from forecast cube	Cache per time/model	Cloudbase, pressure, CAPE thresholds.
Markers	MapLibre symbol/circle layers	Launch/site API	Client cache	Cluster at low zoom, uncluster near launch.
Airspace	Vector polygon/line layers	OpenAIP/national imports	Versioned cache	Must show floor/ceiling labels and active state.
Live tracks	WebSocket + line layer/deck.gl PathLayer	Tracking service	Short-lived memory cache	Low latency in events and rescue mode.
Hazards	Vector polygons + animated edge	Risk engine / playbooks	Cache per site/time	Critical hazards must remain visible above decorative layers.

Map layer stack order

- 00 background
- 10 terrain_hillshade
- 20 weather_scalar_raster
- 30 wind_particles
- 40 weather_isolines
- 50 airspace_polygons
- 60 hazard_zones
- 70 launch_markers
- 80 live_tracks
- 90 labels
- 100 selected_site_callout
- 110 critical_alert_overlay

10. Layer Registry and Entitlement Gating

All layers must be declared in a registry. The registry controls UI labels, API endpoints, legends, units, default opacity, dependencies, entitlement tier and safety behavior. The front end must not hardcode independent layer lists in multiple places.

```
export const weatherLayerRegistry = {
  altitude_wind: {
    label: 'Wind at Altitude',
    icon: 'wind',
    tier: 'PRO',
    endpoint: '/v1/map/weather/altitude-wind/tiles/{z}/{x}/{y}',
    params: ['modelRunId', 'time', 'altitudeM'],
    units: 'km/h',
    defaultOpacity: 0.78,
    legend: 'windRamp',
    dependencies: ['forecast_cube.u', 'forecast_cube.v'],
  },
}
```

```

safetyImpact: ['drift', 'xc', 'rotor']
},
foehn: {
  label: 'Foehn Indicator',
  icon: 'foehn',
  tier: 'PRO_PLUS',
  endpoint: '/v1/map/risk/foehn/tiles/{z}/{x}/{y}',
  params: ['modelRunId', 'time'],
  units: 'risk 0-100',
  defaultOpacity: 0.62,
  legend: 'riskRamp',
  dependencies: ['pressure_gradient', 'cross_barrier_wind', 'humidity_drop'],
  safetyImpact: ['hard_blocker_possible']
}
} as const;

```

Entitlement state	UI behavior	API behavior
Available	Layer toggle enabled; legend visible when active.	Request includes bearer token; normal response.
Locked by tier	Layer row visible with lock/pro badge; click opens upgrade sheet.	API returns 402/403 entitlement error if manually requested.
Unavailable for region	Layer row disabled with reason: no data source in region.	API returns 204 or metadata no_data.
Data stale	Layer active with stale badge; decision engine may downgrade confidence.	Response contains quality flags.
Safety critical	Layer cannot be hidden from NO-GO explanation cards.	Decision endpoint includes hard blocker evidence.

11. Front-End State Architecture

Weather map state must be shareable, recoverable and predictable. The state model is split into URL state, server state, local UI state and offline persisted state.

State class	Examples	Storage	Owner
URL state	selected site, active layer, altitude, time, model, viewport	Query string / hash	Router and map controller.
Server state	forecast responses, site details, entitlements, alerts	TanStack Query/SWR cache or equivalent	API client layer.
Local UI state	panel open/closed, hover marker, legend expanded	Client store/context	UI shell.
Offline state	favorite sites, downloaded forecasts, map packs, last known profile	IndexedDB/mobile database	Offline sync manager.
Safety state	in-flight active, last telemetry, unresolved critical alert	Local durable store + server sync	Safety assistant.

Example URL

```

/map?lat=45.82&lon=11.80&zoom=8.2&site=montegrappa&layer=altitude_wind&altitude=1000&time=2026-06-05T12:00Z&model=ecmwf

```

12. API Data Contracts Used by the Front End

```
GET /v1/sites/{siteId}/decision?time=...&profile_id=...
{
  "site_id": "montegrappa",
  "status": "GO",
  "flyability_score": 82,
  "risk_score": 26,
  "confidence": 0.92,
  "best_window": {"start": "11:00", "end": "15:00", "timezone": "Europe/Rome"},
  "metrics": {
    "wind_launch_kmh": 12,
    "wind_direction": "NW",
    "gust_kmh": 24,
    "cloudbase_m_agl": 1950,
    "thermal_strength_ms": 3.2,
    "cape_jkg": 180
  },
  "blockers": [],
  "warnings": ["Sea-breeze component later in afternoon"],
  "evidence": ["forecast_cube:ecmwf:2026-06-05T00Z", "site_playbook:v4"]
}

GET /v1/map/layers
{
  "layers": [
    {"id": "surface_wind", "label": "Surface Wind", "tier": "FREE", "available": true},
    {"id": "altitude_wind", "label": "Wind at Altitude", "tier": "PRO", "available": true},
    {"id": "foehn", "label": "Foehn Indicator", "tier": "PRO_PLUS", "available": false, "reason": "upgrade_required"}
  ]
}
```

13. Desktop Layout Specification

Region	Default size	Content	Notes
Left sidebar	280 px	Brand, search, favorites, quick layers, drive card	Collapsible to icon rail at 1024-1200 px.
Map viewport	Fluid	Map, overlays, markers, airspace, floating controls	Must not be blocked by non-critical panels.
Right detail panel	420-460 px	Selected launch, decision, hourly table, AI briefing	Resizable in future; closed state uses site callout.
Bottom timeline	160-220 px	Date selector, play button, time scrubber, variable strips	Can collapse to 64 px compact timeline.
Top legend	Floating centered	Layer legend, units, scale	Must update when active layer changes.

- Keyboard shortcut Cmd/Ctrl+K opens search command palette.
- Esc closes temporary overlays but must not dismiss critical safety alerts.
- Clicking a launch marker opens the site decision panel and updates the URL.
- Dragging the timeline updates map tile requests and site decision data with debounce.
- Layer toggles should not reset selected site or timeline state.

14. Mobile and Field UX Specification

Mobile UX must support glare, cold fingers, gloves, poor connectivity and fast decision-making at launch. Dense meteorological detail is secondary to the decision state and hard blockers.

Screen	Mobile layout	Primary actions	Offline behavior
Map home	Full-screen map, top search, bottom selected-site sheet	Search, locate me, layers, favorites	Show cached sites and last forecast with stale warning.
Site sheet compact	GO/MAYBE/NO-GO, wind, gust, best window	Expand, favorite, navigate, alert	Use cached decision if fresh enough; otherwise UNKNOWN.
Layer drawer	Bottom drawer with quick toggles and altitude selector	Toggle layers, change altitude, upgrade	Hide unavailable data, preserve chosen layer.
Timeline	Horizontal scrubber in sheet	Scrub time, play, switch day	Cache 24-72h if user downloaded site pack.
In-flight mode	Large HUD, audio alerts, airspace, glide safety	Start/stop track, SOS, mark landing	On-device rules continue without network.
AI briefing	Expandable grounded summary	Ask follow-up, copy/share	Disabled if data provenance missing.

15. Accessibility and Readability

- All GO/MAYBE/NO-GO states require text labels and icons; never rely on color alone.
- Minimum tap target: 44 x 44 px on mobile; in-flight controls should be 56 x 56 px or larger.
- Critical alerts must support screen readers and haptic/audio channels.
- Weather legends must include units, min/max values and no-data indicator.
- Dark panels must maintain readable contrast over bright map areas using glass/blur plus solid fallback.
- Tables must be horizontally scrollable on mobile without truncating important safety rows.
- The app must support reduced motion; wind particles can be paused or replaced by arrows.

16. Performance Budgets

Metric	Target	Failure behavior
Initial map shell render	< 2.5 s on mid-tier mobile network	Show skeleton and cached base map.
Layer toggle visual response	< 150 ms local response	Show loading shimmer while tile requests continue.
Forecast decision fetch	p95 < 800 ms from API cache	Use stale-while-revalidate card if cache exists.
Map pan frame rate	>= 45 fps desktop, >= 30 fps mobile	Reduce particle density and disable heavy overlays.
Particle layer memory	< 256 MB desktop, < 96 MB mobile	Auto switch to static arrows.
Timeline scrub update	Debounced 150-300 ms	Cancel prior tile requests and preserve last stable view.
Offline site pack size	< 250 MB per region target	Warn before cellular download.

17. Front-End Package Architecture

```

repo/
  apps/
    web/          # Next.js or Vite React web app
    mobile/       # Flutter or React Native app
    storybook/    # component documentation and visual tests
  packages/
    design-tokens/ # token source + CSS/Flutter exports
    ui-react/      # web components
    ui-flutter/    # mobile component mapping
    map-core/      # layer registry, legends, map adapters
    api-client/    # typed API client from OpenAPI
    weather-models/ # shared TypeScript domain types
    test-fixtures/ # golden forecast/site samples

```

Package	Responsibility	Must not contain
design-tokens	Source of color, type, spacing, radius, elevation, motion tokens.	Business logic or API calls.
ui-react	Reusable React components and composition patterns.	Map engine-specific code except adapter components.
map-core	Layer registry, legend configs, tile URL builder, map event contracts.	Authentication or billing code.
api-client	Typed OpenAPI client, request headers, retry policy.	UI state or components.
weather-models	Shared DTO/domain types for forecast, decision, site, tracks.	Runtime rendering code.
test-fixtures	Golden datasets for visual and logic tests.	Production secrets or user data.

18. Component States and Empty States

State	When it appears	Required UI behavior
Loading	API request pending, tile loading, first view	Skeleton or map-preserving spinner; do not clear previous decision unless outdated.
Stale	Forecast/model run older than freshness threshold	Show stale badge and downgrade confidence.
No data	No forecast or layer for region/time	Disable decision; show UNKNOWN, not GO.
Partial data	Some variables missing	Show available metrics and list missing blockers.
Locked	Feature requires paid tier	Show preview, lock badge and upgrade explanation without hiding the feature completely.
Error	API or render failure	Human-readable recovery action; developer error code in details.
Critical	Safety hazard detected	Interruptive banner; do not hide under drawers or modals.

19. Storybook and Visual Regression

Every major component should have stories for normal, loading, error, stale, locked, no-data and critical states. Visual regression must include map panels over bright terrain and dark terrain backgrounds, because contrast problems often appear only when panels overlay complex maps.

Storybook story matrix

SiteDecisionPanel

- GO / high confidence
- MAYBE / model disagreement
- NO_GO / hard blocker: gusts
- UNKNOWN / missing cloudbase
- Locked AI briefing
- Mobile bottom-sheet variant
- Dark terrain background
- Bright snow/terrain background

20. Design System Governance

- Every new component requires a component owner, Storybook examples, accessibility notes and test fixture.
- Weather color ramps require review by product, weather-engine and accessibility owners before release.
- Safety wording changes require safety/legal review because they can alter pilot behavior.
- Layer registry changes require API, weather-engine and front-end review.
- Paid-tier gates must use the entitlement service, not hardcoded frontend booleans.
- Deprecated components remain available for one minor release with warnings.

21. Implementation Examples

```
// Weather layer toggle flow
function onToggleLayer(layerId: WeatherLayerId) {
  const layer = weatherLayerRegistry[layerId];
  if (!entitlements.canUse(layer.tier)) {
    openUpgradeSheet({ feature: layer.label, requiredTier: layer.tier });
    return;
  }
  mapStore.toggleLayer(layerId);
  router.setQuery({ layer: mapStore.activeLayers.join(',') });
  analytics.track('weather_layer_toggled', { layerId, tier: layer.tier });
}

// Map tile request builder
export function weatherTileUrl(layer: LayerConfig, ctx: LayerRequestContext) {
  const params = new URLSearchParams({
    model_run_id: ctx.modelRunId,
    time: ctx.timeISO,
    opacity: String(ctx.opacity)
  });
  if (ctx.altitudeM) params.set('altitude_m', String(ctx.altitudeM));
  return `${API_BASE}${layer.endpoint}?${params.toString()}`;
}
```

22. Front-End Security and Privacy Controls

Risk	Control
Live location exposure	Default private tracking; explicit share links; expiry and

	revoke controls.
Entitlement bypass	Backend enforces tiers; frontend only improves UX.
XSS in community reports	Escape all text; sanitize markdown; disallow arbitrary HTML.
Token leakage	Use secure storage; do not log Authorization headers; rotate refresh tokens.
AI prompt injection through notes/reports	Treat community and playbook text as untrusted data; AI service handles grounding and quoting.
Map tile URL leakage	Signed or scoped URLs for paid/restricted layers; avoid secrets in query strings.
Screenshot privacy	In-flight/live tracking screens include privacy mode and hide exact coordinates when needed.

23. Product Analytics Events

Event	When fired	Key properties
map_view_loaded	Map shell becomes interactive	viewport, device, tier, cached
site_selected	User selects launch marker or favorite	site_id, source, distance_km
decision_viewed	GO/MAYBE/NO-GO card displayed	site_id, state, score, risk, confidence
weather_layer_toggled	User toggles map layer	layer_id, tier_required, active_count
altitude_changed	Altitude wind selector changes	altitude_m, layer_id
timeline_scrubbed	User changes forecast time	delta_hours, site_id, layer_id
ai_briefing_opened	User expands AI card	site_id, tier, confidence
upgrade_prompt_clicked	Locked feature prompts upgrade	feature_id, required_tier
alert_rule_created	User creates alert rule	site_id, rule_type, channel
inflight_mode_started	Pilot starts in-flight mode	offline_available, tracking_enabled

24. QA Acceptance Criteria

- All primary map layers are selectable from the layer panel and represented in the layer registry.
- The selected layer, time, altitude and site are reflected in the URL and restored on reload.
- Locked paid layers show upgrade affordances but cannot fetch restricted data without backend authorization.
- GO/MAYBE/NO-GO states remain readable over all tested map backgrounds.
- Mobile site sheet can be used one-handed and does not hide critical alerts.
- Timeline scrubber updates the map and decision panel without race-condition flicker.
- Reduced-motion mode disables animated particles and preserves information through arrows/rasters.
- Offline cached site forecast displays staleness and never silently presents old data as current.
- Storybook contains normal, loading, error, no-data, stale and locked states for every major component.
- Visual regression tests include desktop, tablet and mobile breakpoints.

25. Front-End Roadmap

Phase	Goal	Deliverables
MVP	Prove map-first launch decision workflow	Map shell, search, favorites, 6 layers, decision panel, timeline, site details, basic alerts.
V1	Commercial Pro app	Altitude wind, AI briefing, paid-tier locks, offline site packs, mobile

		bottom sheets.
V2	Competition and professional portals	Event weather center, club/school portal UI, live tracking dashboard, moderation console.
V3	Advanced safety and digital twin UI	Rotor/Foehn visualizations, launch playbooks, forecast-vs-reality, expert validation panels.
V4	Global polish and ecosystem	Wearables, widgets, embedded club maps, white-label dashboards, advanced accessibility personalization.

26. Engineering Backlog

Epic	Priority	Stories
Design token pipeline	P0	Create token JSON, CSS variables, Flutter export, Storybook theme switch.
Map shell	P0	Implement AppShell, MapViewport, layer registry and URL state sync.
Weather layers	P0	Surface wind, altitude wind, gusts, cloudbase, thermals and radar layers.
Decision panel	P0	Flyability/risk cards, hourly table, launch window and blocker tree.
Timeline	P0	Scrubber, play mode, hourly variable strips and debounced layer refresh.
Mobile UX	P1	Bottom sheet, layer drawer, offline site pack UI and compact GO/MAYBE/NO-GO cards.
AI briefing UI	P1	Grounded AI card, provenance link, confidence display and beta label.
Entitlements	P1	Locked layer states, upgrade sheet, subscription status badge.
Visual regression	P1	Storybook stories and Playwright screenshot baselines.
In-flight UI	P2	Safety HUD, airspace alert banner, SOS share sheet and offline warnings.

27. Technical References

- MapLibre GL JS documentation - WebGL vector maps and style-driven rendering: <https://www.maplibre.org/maplibre-gl-js/docs/>
- deck.gl MapLibre integration documentation - interleaved rendering with MapLibre WebGL context: <https://deck.gl/docs/developer-guide/base-maps/using-with-maplibre>
- Flutter adaptive and responsive design documentation: <https://docs.flutter.dev/ui/adaptive-responsive>
- Material Design 3 design system documentation: <https://m3.material.io/>
- MapLibre Style Specification expressions: <https://www.maplibre.org/maplibre-style-spec/expressions/>

Appendix A - Feature to UI Surface Matrix

Feature	Desktop surface	Mobile surface	Criticality
Launch search	Left sidebar / command palette	Top search / command palette	High
Layer selection	Left quick layers + advanced layer modal	Layer drawer	High
Altitude wind	Floating altitude selector	Layer drawer segmented control	High
GO/MAYBE/NO-GO	Right panel + map callout	Bottom sheet compact card	Critical
Hourly forecast	Right panel table + bottom strips	Scrollable table/sheet	High
AI briefing	Right panel card	Expandable bottom-sheet card	Medium
Drive decision	Left card	Site sheet / notification	Medium
Airspace	Map overlay + toggle	Map overlay + toggle	Critical in-flight
In-flight alert	Top/center critical banner	Full-width critical banner + voice	Critical
SOS/rescue	In-flight mode	In-flight mode	Critical

Appendix B - Front-End Definition of Done

- Component implemented with typed props and no hardcoded feature entitlement logic.
- Component has Storybook stories for standard state variants.
- Component passes accessibility checks and keyboard navigation where applicable.
- Component uses design tokens rather than hardcoded colors/spacing.
- API integration has loading, error, stale and empty states.
- Visual regression baseline is approved at desktop/tablet/mobile widths.
- Safety wording matches the approved wording dictionary.
- Analytics event is emitted for business-critical interactions.
- Documentation includes component purpose, props and state chart.